National College of Ireland

Project Submission Sheet

Student Name:	RAGHUL SURESHBABU		
Student ID:	23377755		
Programme:	MSc Data Analytics	**Year:**	2025
Module:	Data Mining and Machine Learning (MSCDAD_A_JAN25I)		
Lecturer:	Jaswinder Singh		
Submission Due Date:	Monday, 5th May 2025 @ 23.55 Hours		
Project Title:	DATA – DRIVEN MODELLING OF STUDENT ACADEMIC PERFORMANCE AND FINANCIAL DEBT IN HIGHER EDUCATION		
Word Count:	3468		

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

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Signature:	Raghul Sureshbabu
Date:	05-05-2025

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1. Please attach a completed copy of this sheet to each project (including multiple copies).
2. Projects should be submitted to your Programme Coordinator.
3. **You must ensure that you retain a HARD COPY of ALL projects**, both for your own reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specifically requested.
4. You must ensure that all projects are submitted to your Programme Coordinator on or before the required submission date. **Late submissions will incur penalties.**
5. All projects must be submitted and passed in order to successfully complete the year. **Any project/assignment not submitted will be marked as a fail.**

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AI Acknowledgement Supplement

I. MODULE: DATA MINING AND MACHINE LEARNING (MSCDAD_A_JAN25I)

II. TITLE: DATA-DRIVEN MODELLING OF STUDENT ACADEMIC PERFORMANCE AND FINANCIAL DEBT IN HIGHER EDUCATION

Your Name/Student Number	Course	Date
RAGHUL SURESHBABU - 23377755	**Data Mining and Machine Learning (MSCDAD_A_JAN25I)**	**05-05-2025**

This section is a supplement to the main assignment, to be used if AI was used in any capacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click here.

III. AI ACKNOWLEDGMENT

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
NA	NA	NA

IV. DESCRIPTION OF AI USAGE

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used**.

NA	
NA	
NA	NA

V. EVIDENCE OF AI USAGE

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

VI. ADDITIONAL EVIDENCE: NA

[Place evidence here]

DATA – DRIVEN MODELLING OF STUDENT ACADEMIC PERFORMANCE AND FINANCIAL DEBT IN HIGHER EDUCATION

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***Abstract*—This research employs supervised Machine Learning to analyze and predict important academic performance through genuine data collected through higher education institutions. The Open University Learning Analytics repository contains the first dataset that enables classification of academic final results between Pass and Fail and Distinction and Withdrawn through analyzing demographic and behavioral features like virtual learning environment activity and assessment score averages. The data from the U.S. Department of Education allows researchers to apply regression methods for predicting student debt amounts at graduation by examining financial, admission and institutional factors. The study utilized Random Forest together with Logistic Regression as classification algorithms while Random Forest and Gradient Boosting Regressors tested for debt prediction. Precision, recall and accuracy and R^2 served as the evaluation metrics to assess the developed models. Total clicks with average score possess high predictive power for academic success but admission rate together with tuition fees provide significant insights into debt prediction. Through this work researchers show how educational institutions can benefit from machine learning for obtaining practical insights.**

1. INTRODUCTION

Education system had a major digital transformation which provides with the extensive student data that gave enhanced insights regarding academic performances of the students and the institutional choices. The predictive patterns of educational data through machine learning methods enabled strong predictive innovations while supporting resource optimization and improving educational policies. The study focuses on two important educational research goals including student academic prediction through behavioral & demographic data and student debt estimation by forecasting graduation debts.

The dual-purpose analysis receives support through two extensive datasets which include Open University Learning Analytics for classification (Dataset1) work alongside U.S. Department of Education College Scorecard for regression examinations (Dataset 2). This study aims to solve two research questions:

- Do learning behavior and personal attributes provide satisfactory methods for assigning student end results?
- Does the combination of educational establishment attributes and social economic measures help forecast student debt amounts?

The analysis utilized Logistic Regression and Random Forest (Dataset 1) along with Random Forest Regressor and Gradient Boosting Regressor (Dataset 2) to resolve these research questions. The study analyzed outcomes using suitable performance metrics which included classification accuracy as well as R-squared (R^2) and Mean Squared Error (MSE).

The document unfolds into the following sections:

- Section 2: The review section of the document examines both research papers and methods within educational data mining.
- Section 3: Justification of the selected machine learning models and their association with datasets.
- Section 4: This section details the experimental approach from data preprocessing through model training and evaluation plan.
- Section 5: The research paper explores interpretation methods alongside performance outcomes of the models.
- Section 6: Final points and future work will be presented

2. LITERATURE REVIEW

The implementation of machine learning methods has expanded remarkably throughout the previous years for predicting educational results and forecasting higher education funding needs. This section analyzes essential research with methodological approaches from the field to guide the selection of methods used in this research project.

a. Student performance prediction

Study results have investigated both behavioral data alongside demographic variables in student performance prediction. Support Vector Machines proved their ability to predict Massive Open Online Course dropout behaviors based on clickstream data according to Kloft et al. (2014).. Random Forest and Naive Bayes classifiers proved successful in detection of failure risk among students through integrating student engagement metrics with academic history records according to Lykourentzou et al. (2009). The Open University dataset captures information about virtual learning environment (VLE) interactions, age, disability status and previous education level while fitting into established research studies and serving as a benchmark model for classification assessment.

b. Educational data mining techniques

According to Romero and Ventura (2010) the preprocessing of educational data remains vital especially for handling three key elements: missing values, categorical encoding and class imbalance. Within educational contexts the applied baseline model uses Logistic Regression due to its powerful capability for interpreting and analyzing categorical variables. The Random Forest ensemble model displays higher capability in identifying nonlinear relationships while performing better with imbalanced datasets according to Bhardwaj and Pal (2012).

c. Previous uses of this dataset

Academic and industrial research makes extensive use of the Open University dataset to provide benchmark standards for detecting dropouts and student classification models. The College Scorecard dataset operated by the U.S. Department of Education serves economic policy researchers to study student outcome effects from institution costs.

d. Gaps and contribution

Some studies investigate academic performance and student debt as separate problems but very few adopt a combined strategy that links student behaviors with financial considerations in higher education. Research presented in this study adds to academic literature through its integration of final result prediction (classification) with debt estimation (regression analysis). This project uses log transformations by filtering the outliers and hyperparameter tuning to build a robust model for this model development approach.

3. CHOICE OF METHODS

The selection process of machine learning algorithms for this project combined prediction task nature with dataset characteristics and accuracy vs interpretability requirements fulfilment

3.1 Classification – Dataset 1 (Student Final Result Prediction):

The Open University Learning Analytics Dataset contains student demographic information expressed through numbered and identified variables which include characteristics such as student age band as well as gender and educational attainment alongside behavioural measures such as total Virtual Learning Environment clicks and average assignment scores. The goal of this task involves predicting student categorization between Pass, Fail, Withdrawn, and Distinction.

The following models were deployed to handle this problem which consists of multiple classification categories.

a. Logistic Regression:

Logistic Regression establishes itself as a solid foundation because it keeps things simple in addition to being easily understood. This model uses linear relationships to link its predictor variables with log-odds of belonging to a particular class. The model functions well as a tool to identify core student outcome predictors while lacking capability in non-linear pattern detection.

b. Random Forest

The Random Forest algorithm produced appropriate outcomes because it combines the capability to handle datasets with varied variable types along with feature relationships detection. The prediction method uses decision trees ensemble for selection and reduces errors while allowing users to evaluate feature importance for interpretation needs. Class balancing methods along with optimized hyperparameters produce superior classification outcomes when used together because this approach recommends this model for handling imbalanced educational datasets.

The projected modeling structure meets predication requirements while providing a system that reveals student intervention events.

3,2 Regression – Dataset 2 (Student Debt Prediction)

The U.S. College Scorecard dataset contains institutional and financial features about admission rates, in-state tuition fees and cost of attendance together with Pell Grant recipient percentages to predict the median student debt at graduation.

The models presented the following regressions for predicting this continuous target value.

A. Random Forest Regressor:

The choice of Random Forest regression became necessary due to its exceptional capabilities in dealing with high-dimensional non-linear problems and its resistance to outlier values. The model derives benefits from financial data analysis because it does not require distribution specifications. The model has an improved capability to detect interrelationships between features than standard linear prediction systems..

B. Gradient Boosting Regressor:

The decision focused on Gradient Boosting as an advanced ensemble technique that creates multiple trees in sequence which refine the existing residuals step by step. The model produces enhanced predictive power than Random Forest models when appropriately adjusted yet stands out for its ability to identify intricate non-linear associations. This method served to extend the established baseline performance of Random Forest.

The regression models performed better through log transformations of skewed data features as well as Z-score filtering for outlier removal to achieve improved steady model performance and accurate predictions.

4. METHODOLOGY

4.1 Business Understanding

Dataset 1:

The objective of Dataset 1 is to predict final academic module results of open university students through statistical analysis of demographic and behavioural data with assessment results. Educational institutions boost their support interventions through identifying relationships between academic results and student achievement patterns to supply early assistance when required as well as personalise learning techniques. The student classification task divides participants into the categories 'Pass' 'Fail' 'Withdrawn' 'Distinction'. The predictive analytics help educational institutions keep students enrolled longer and monitor progress through dashboard monitoring systems which direct support measures towards students needing help.

Dataset 2:

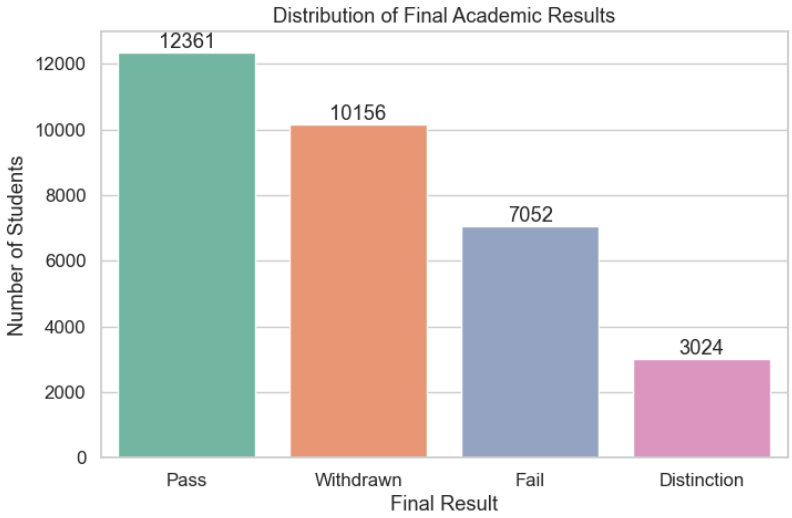
The objective of Dataset 2 is to forecast median student debt amounts called DEBT_MDN through various institutional, financial, and demographic variables. Student debt patterns research gives officials and educational organisations and prospective students the information needed to create lasting financial plans that support the accountability of institutions. The regression task predicts future values and conducts longitudinal studies across different colleges to aid better financial understanding in education.

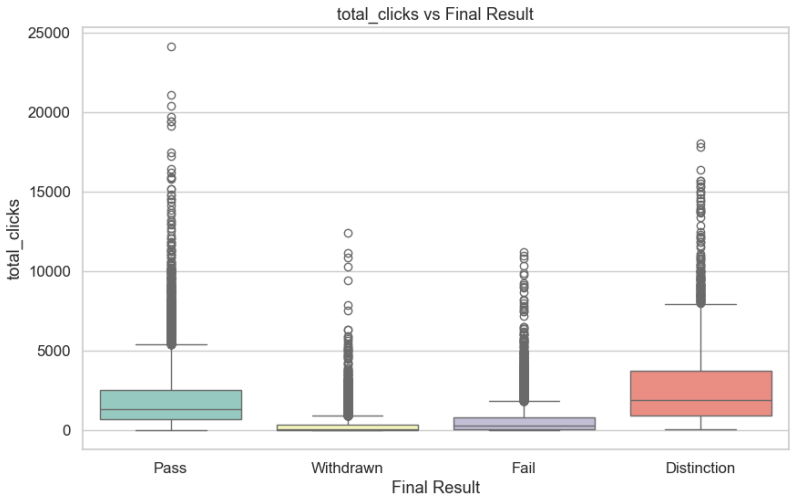
4.2 Data Understanding

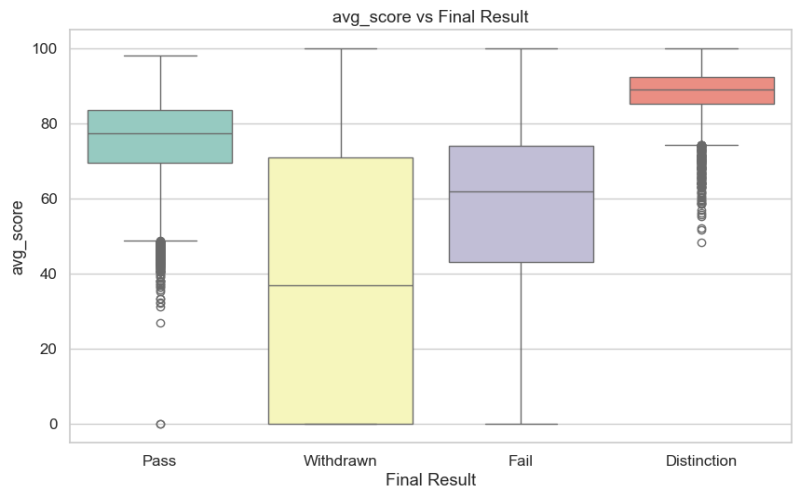
Dataset 1:

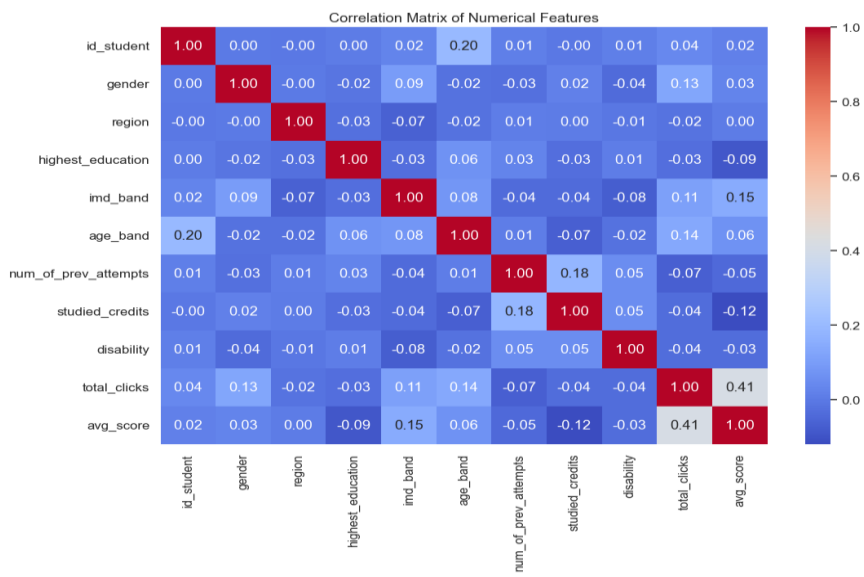
The first dataset was created by merging studentInfo with studentAssessment as well as studentVle records obtained from the Open University Learning Analytics database. The cleaned and prepared data included 32,593 records. The dataset consists of categorical information including gender, region, highest_education and imd_band and age_band and disability variables while numerical data contains total_clicks and avg_score measurements. EDAs performed an initial analysis to verify minimal missing information together with visualising the feature distributions while revealing the extreme class imbalance through 'Pass' observations being dominant. To cheque for variable interdependence heatmaps combined with correlation plots were used.

Heatmap of Missing Values in Dataset
0
1208
2416
3624
4832
6040
7248
8456
9664
10872
12080
13288
14496
15704
16912
18120
19328
20536
21744
22952
24160
25368
26576
27784
28992
30200
31408
code_module
code_presentation
id_student
gender
region
highest_education
imd_band
age_band
num_of_prev_attempts
studied_credits
disability
final_result
total_clicks
avg_score


Distribution of Final Academic Results
12361
10156
7052
3024
Number of Students
0
2000
4000
6000
8000
10000
12000
Pass
Withdrawn
Fail
Distinction
Final Result

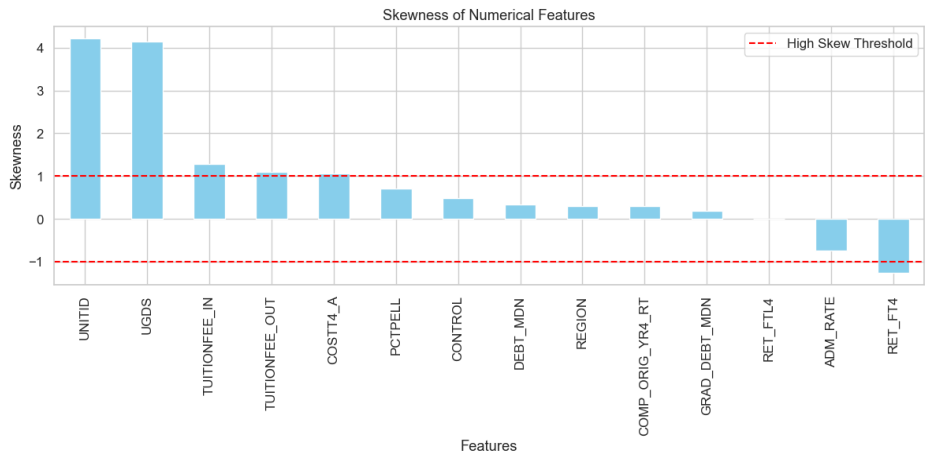

total_clicks vs Final Result
total_clicks
0
5000
10000
15000
20000
25000
Pass
Withdrawn
Fail
Distinction
Final Result

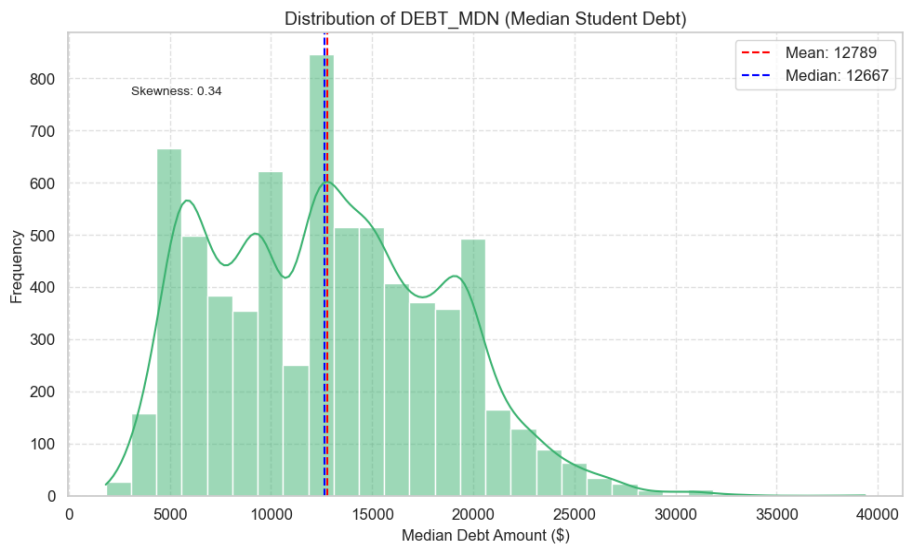


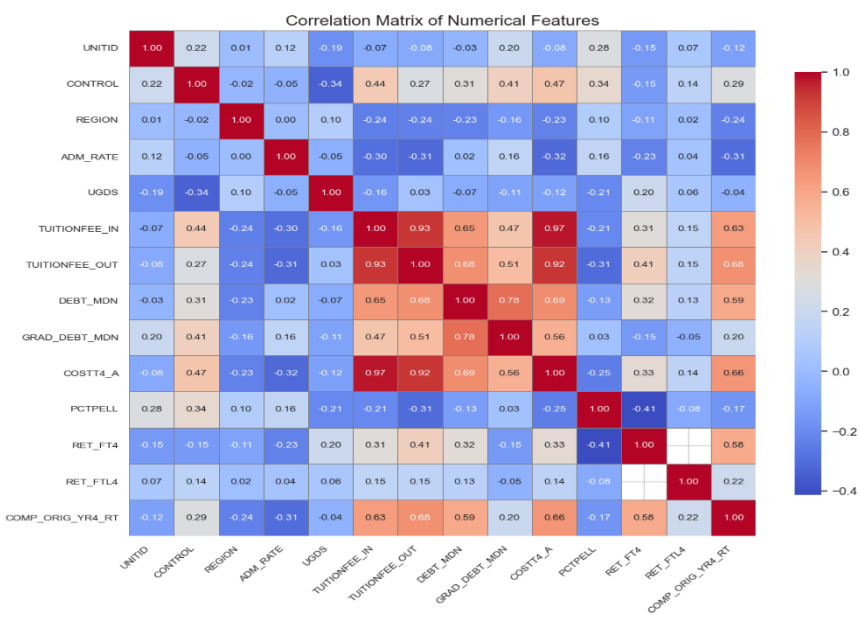


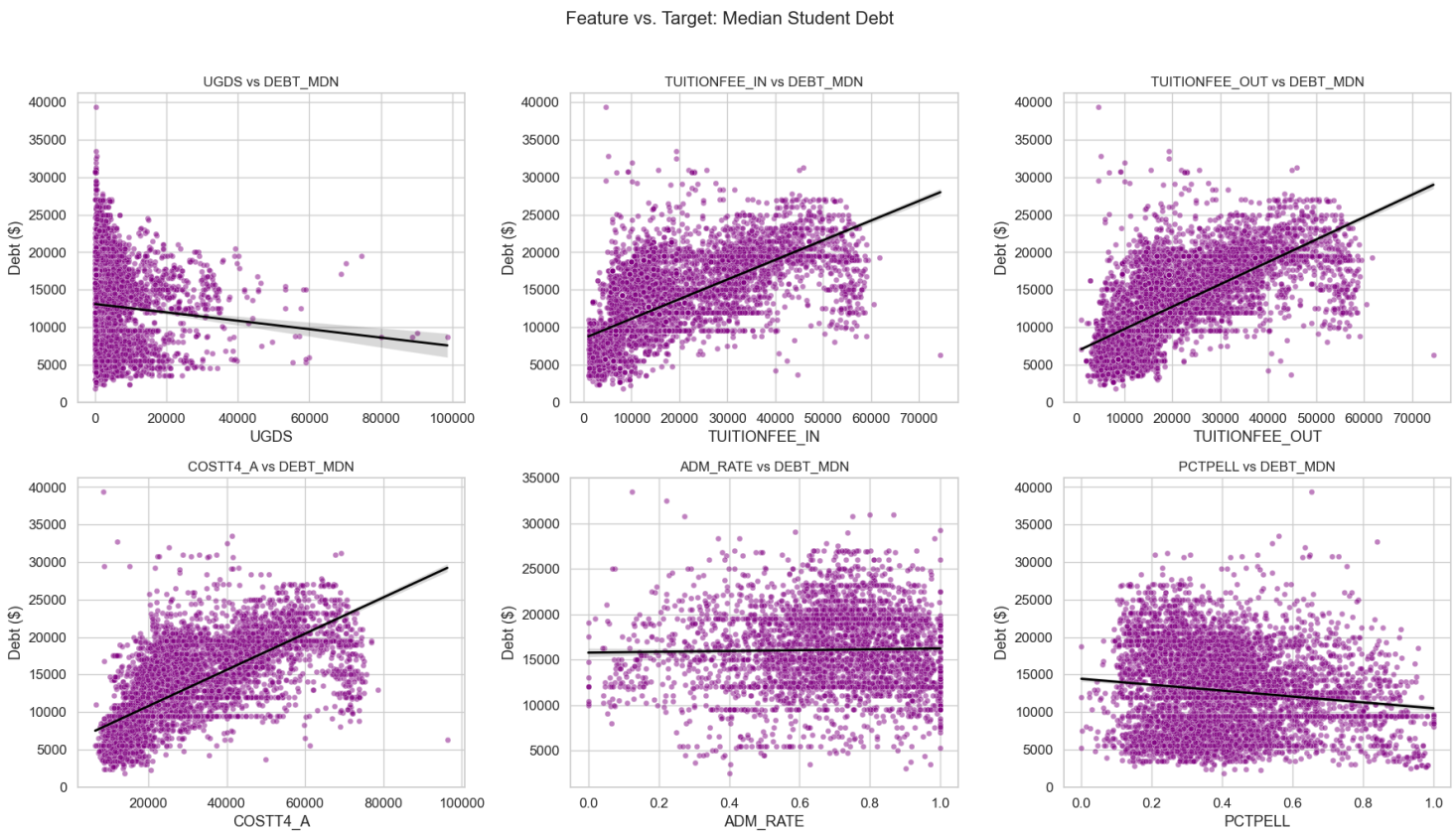
Dataset 2:

Over 6,000 valid records remained after cleaning and filtering the data which originated from the U.S. Department of Education College Scorecard. The study examines six principal factors which consist of undergraduate enrollment (UGDS), tuition (in-state and out-of-state), cost of attendance (COSTT4_A), admission rate (ADM_RATE), Pell Grant recipient percentage (PCTPELL) and control type (CONTROL). The analysis of college data included visual inspections through histograms combined with KDE plots along with scatter plots to produce a correlation matrix. Higher values of the DEBT_MDN variable produced positive skewness and wide institutional variation thus requiring transformation later in modelling stages. A significant number of analysed features exhibited distribution irregularities combined with statistical correlation among their values. The results from scatter plot analysis demonstrated strong positive relationships between DEBT_MDN values and all tuition fee components (such as TUITIONFEE_IN and TUITIONFEE_OUT as well as COSTT4_A) yet correlations with UGDS and PCTPELL were minimal. Visual relationships provided evidence which supported the use of these predictors during data preprocessing through log transformation and scaling methods.


Skewness of Numerical Features
High Skew Threshold
Skewness
Features
UNITID
UGDS
TUITIONFEE_IN
TUITIONFEE_OUT
COSTT4_A
PCTPELL
CONTROL
DEBT_MDN
REGION
COMP_ORIG_YR4_RT
GRAD_DEBT_MDN
RET_FTL4
ADM_RATE
RET_FT4


Distribution of DEBT_MDN (Median Student Debt)
Mean: 12789
Median: 12667
Skewness: 0.34
Frequency
Median Debt Amount ($)


Correlation Matrix of Numerical Features
UNITID
CONTROL
REGION
ADM_RATE
UGDS
TUITIONFEE_IN
TUITIONFEE_OUT
DEBT_MDN
GRAD_DEBT_MDN
COSTT4_A
PCTPELL
RET_FT4
RET_FTL4
COMP_ORIG_YR4_RT



4.3 Data Preparation

Dataset 1:

To denote students who did not participate in online activities the imputation method set total_clicks and avg_score at zero. The application of label encoding allowed the transformation of categorical attributes into numbers which became suitable for use in classification models. The predominantly tree-based Random Forest implementation did not require feature scaling since the analysis depended on observing student behaviour variations in their natural state. The train-test split contained stratification which protected the distribution of class labels.

Dataset 2:

The analysed features underwent numeric conversion then NA value management before log1p transformation normalised the four distributionally-skewed variables (UGDS, TUITIONFEE_IN, TUITIONFEE_OUT, COSTT4_A). A Z-score filtering method (threshold < 3) was utilised to eliminate outliers because they could distort the regression outputs. StandardScaler scaling was applied to the features so the model could maintain stability. The processing pipeline helped to minimise bias when making predictions while enhancing the convergence stability of tree-based regression models.

4.4 Modelling

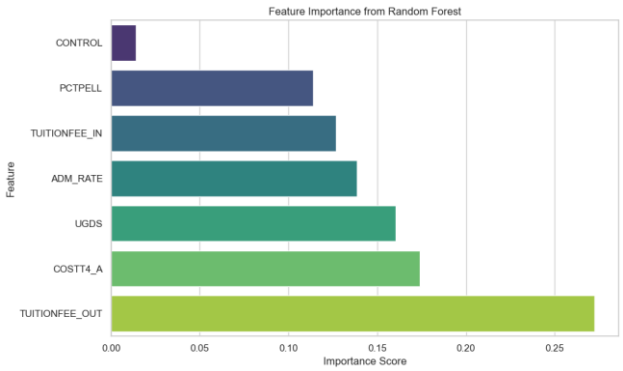
Dataset 1:

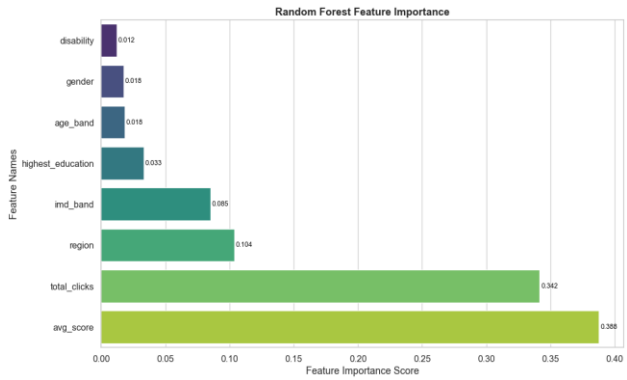
Two classification models were applied to Dataset 1 in order to forecast final academic results. The basic Logistic Regression model achieved 61.7% accuracy but researchers could understand its basic structure. According to the model the classes 'Pass' and 'Distinction' were most accurately predicted but the 'Withdrawn' category was less accurate. While Random Forest Classifier exceeded Logistic Regression regarding recall performance it did not reach above 60.1% for predicting final academic outcomes. The key factors that influence results in predictions emerge from Random Forest through its analysis of vital features. A stratified

split method confirmed the validity of the selected models by optimising performance through GridSearchCV which adjusted the max_depth and n_estimators parameters.

Dataset 2:

The development process generated two regression prediction models through Random Forest Regressor and Gradient Boosting Regressor to forecast DEBT_MDN values from Dataset 2 indicators. By implementing both feature transformation alongside hyperparameter optimization Random Forest achieved a stabilised prediction of student debt with an R^2 value of 0.51.. Although it returned an R^2 score of 0.46 the Gradient Boosting model demonstrated superior generalisation since its residual analysis indicated even distribution across the entire range. The achievement of model convergence along with stability required successive training sessions to use standardised cleaned information from the dataset.. The evaluation process incorporated cross-validation to stop overfitting and determine model stability. Student debt largely relies on tuition fees coupled with cost of attendance and admission rates according to feature importance analysis therefore colleges can create financial strategies and policies through this discovery.





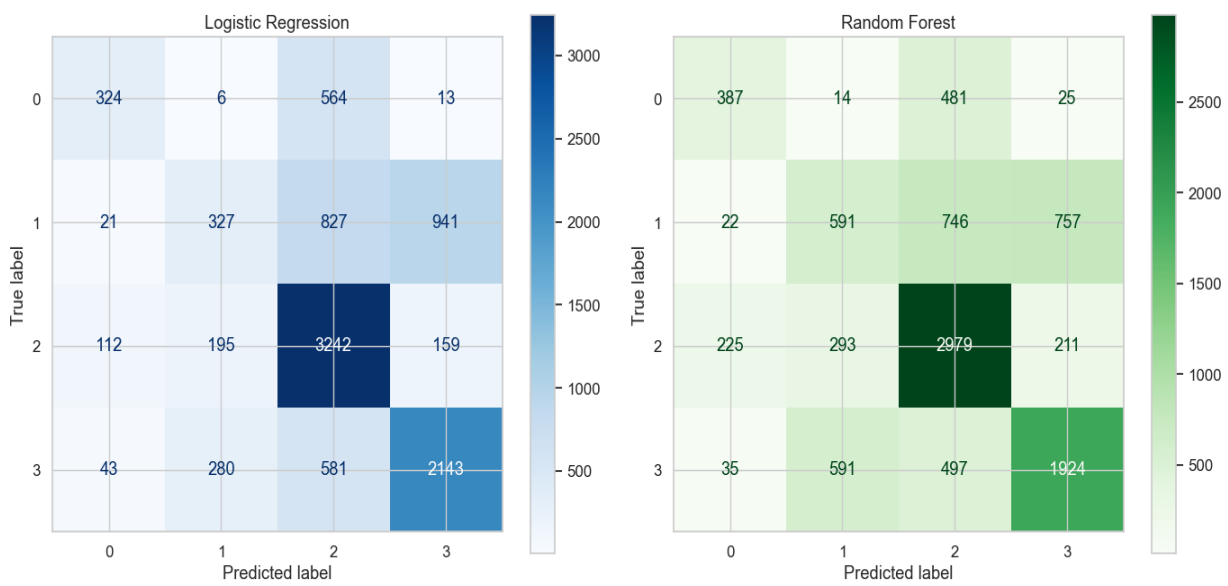
Characteristic	**Dataset 1: Open University (Classification)**	**Dataset 2: College Scorecard (Regression)**
Number of Variables (After Cleaning)	10	12
Records After Cleaning	32,593	7,703
Target Variable	final_result consists of four categories: Pass, Fail, Withdrawn, Distinction	DEBT_MDN
Type of Prediction Task	Classification	Regression
Irrelevant Variables Removed	Yes	Yes
Duplications Removed	Yes	yes
Missing Value Handling	Imputed total_clicks and avg_score with 0	Filled numeric fields with mean; used log1p transformation
Outliers Filtered	No	Yes (Z-score < 3)
Feature Scaling	Not applied since for tree-based models it's not needed	Standard Scaler operated on all numeric features.
Categorical Encoding	Label encoded for all the categorical features	All the features have been converted to numerics
Dimensionality Reduction	No	No
Train-Test Split	70% train / 30% test	70% train / 30% test
Implemented Models	Logistic regression, Random Forest classifier	Random Forest regressor, Gradient Boosting regressor
Hyperparameter Tuning	GridSearchCV (max_depth , n_estimators and learning_rate)	GridSearchCV (max_depth , n_estimators and learning_rate)
Best Performance Metric	Accuracy: 61.7% (LogReg), 60.1% (RF)	R^2: 0.51 (RF), 0.46 (GBR)

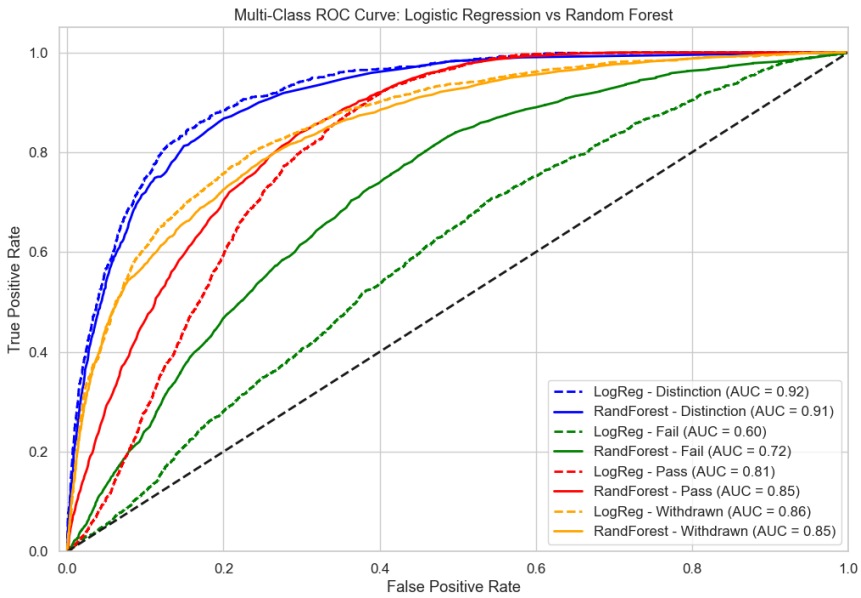
Evaluation Metrics	Accuracy, F1-score, ROC AUC, Confusion Matrix	R^2, MSE, Residual Plots
Feature Importance analysis	Yes (RF)	Yes (RF,GBR)
Cross-Validation	Yes	Yes
Model Interpretability	Moderate (RF), High (LogReg)	High (Feature Importance)

5. EVALUATION

Dataset 1 - Classification

The multiclass classification evaluation utilised accuracy and precision and recall in addition to macro-averaged and weighted F1-score metrics. The Logistic Regression model started with a 61.7% accuracy rating during evaluation which showed positive results particularly for the 'Pass' and 'Distinction' classification categories. The recovery of the 'Withdrawn' class and other rare categories remained poor across the model's performance. The Random Forest Classifier functioned as a solution for class imbalance combined with feature interaction during classification activities. The Random Forest Classifier managed to improve minority class detection by performing better in recall and F1-score evaluations while the accuracy rate decreased to 60.1%. Ensemble learning methods provided better predictive stability toward each student performance class. Confusion matrices demonstrate how the model functions between X and Y variables as Figure X illustrates. The assessment results showed consistent incongruence between students who passed and those who failed resulting in poor separation between Pass and Fail categories within the matrices. Performance interpretation received validation through reports that demonstrated per-class performance metric outputs. The evaluation utilised ROC curve plots at a one-vs-rest strategy that displayed the assessment results in Figure Y. The AUC scores obtained from most of The Logistic Regression model classes surpassed Random Forest scores to produce better separation results. Visual results indicate that Logistic Regression exhibits higher discriminative capabilities while Random Forest shows better recall performance for rare-class prediction because of which it makes an excellent choice for academic risk detection systems.





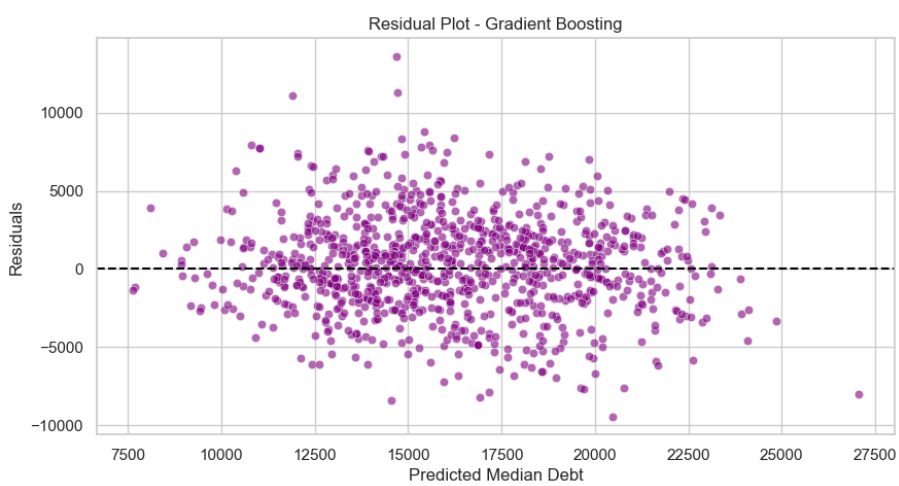
Dataset 2 - Regression

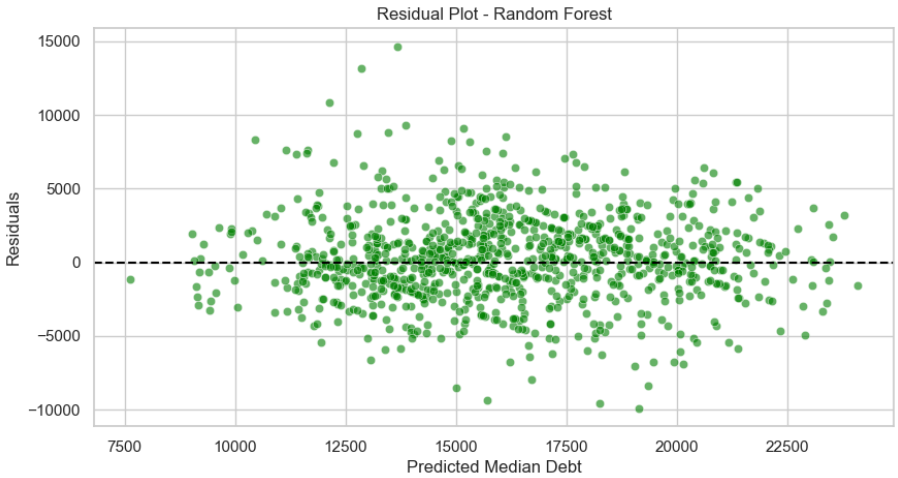
R^2 together with MSE formed the primary metrics which experts used to evaluate regression models. The Random Forest Regressor achieved an R^2 value of 0.51 which explained more than half of the DEBT_MDN student debt changes caused by the analyzed institutional variables and demographic characteristics. Model effectiveness from the Gradient Boosting Regressor seemed slightly decreased with $R^2 \approx 0.46$ but its improved generalisation quality became evident through its superior residual fit.

An examination of regression residuals took place through scatter plots that displayed residuals against predicted values for each of the two regressors as illustrated in Figure Z. The residual values in both plots showed random distribution around zero which validated our homoscedasticity assumption and triggered the removal of systematic modelling errors. The edge regions of Random Forest showed slightly higher residual variance while Gradient Boosting displayed widespread residual distribution that led to better unknown dataset stability.

Using GridSearchCV optimization the models found their optimal max_depth along with pairs of values between n_estimators and learning_rate to maximize performance (for Gradient Boosting). The optimization approach enhanced both the accuracy performance and the generalization capabilities simultaneously. Random Forest generated superior R^2 and more competitive MSE results than Gradient Boosting yet Gradient Boosting demonstrated the most effective performance when assessing institutions with outlier behaviors.

The predictive models generate valid debt projections for students who can be incorporated into educational budgeting platforms for policy evaluation needs.





6. CONCLUSIONS

Dataset 1

The assessment of academic results by machine learning algorithms became possible when organisations leveraged demographic profiles together with engagement behavior and academic records for analysis. Using the Logistic Regression model produced an accuracy baseline of 61.7% through effective identification of the most common outcome categories Pass and Distinction. This linear approach failed to address minority outcomes because of its high sensitivity to minority classes and its linear structure which limited appropriate handling of underrepresented classes such as Withdrawn. The Random Forest Classifier achieved balanced performance results across scenarios and generated accurate recall measures as well as providing knowledge about feature significance. The prediction model identified total_clicks measurement along with avg_score and highest_education variables as key dimensions that determine outcome success. The 60.1% accuracy could not invalidate Random Forest's impressive abilities thus making it an exemplary model for institutional decision systems. The validated analytic models proved that predictive analytics enables institutions to create responsive intervention frameworks that defend students from terminating distance learning programmes.

Dataset 2

Regression studies quantitatively confirmed significant connexions between student debt levels across American higher education institutions. The DEBT_MDN values were effectively predicted by Random Forest and Gradient Boosting Regressors by using institutional and demographic characteristics. The R^2 value reached 0.51 by Random Forest exceeding Gradient Boosting because the model successfully deduced above 50% of debt total differences.. The R^2 value for Gradient Boosting was slightly lower at 0.46 but its stable and generalised prediction patterns became apparent through residual analysis. The main factors that triggered student debt were tuition fees combined with the cost of attendance and admission rates according to results from feature importance analysis which validates existing socioeconomic distribution patterns during higher education financing. These models provide evidence that machines learning systems can effectively support higher education institutions by helping them understand debt patterns as well as plan effectively and make better decisions for debt management and affordability purposes.

7. FUTURE WORK

Dataset 1

- This analysis should review VLE weekly activity patterns because they show the progression of student disengagement behaviour in each period.
- XGBoost and LightGBM should be evaluated by the research team with neural networks to identify optimum accuracy levels in prediction.
- The performance of minority classes needs improvement by applying SMOTE and ADASYN data balancing techniques.
- PCA and feature selection combined methods will enhance the model's generalisation capabilities.
- A feedback mechanism needs to become part of practical academic alert systems to enable instant response capabilities.

Dataset 2

- Several time point data from the investigation help establish changes in student debt patterns.
- Predictions become more accurate by analysing both financial aid records from students and graduate salary and loan payment information after completion of their education..
- A stacking ensemble along with combination models should be tested to extract top performance from Random Forest and Gradient Boosting systems.
- The study depends on propensity score matching methods and instrumental variables to analyse the causal impacts.
- The research must display debt predictions to education finance policy stakeholders through interactive dashboard interfaces as one research component.

8. REFERENCES

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